

When Corrections Backfire: The Impact of Community Notes on MMR Misinformation at the User and Network Levels

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Abstract

MMR misinformation remains a public health concern, and X's Community Notes aim to reduce misleading vaccine claims by adding corrective context. Yet little research has used observational X data to examine within-user changes after exposure, especially across engagement, misinformation sharing, and downstream diffusion. This study compares users' MMR-related posts before and after MMR-related Community Note exposure. The results show that exposure was not associated with significant overall changes in posting activity or misinformation proportion, but users showed significant spikes in both outcomes on the day of exposure, possibly reflecting short-term psychological reactance. A social contagion-based network simulation further suggested that this temporary increase could produce a small but persistent increase in total misinformation diffusion. These findings suggest that Community Notes may fall short of the ideal intervention goal of reducing misinformation while preserving engagement; in this case, visible correction appeared to trigger a short-term backfire pattern that increased misinformation sharing and modestly amplified diffusion.

Introduction

Misinformation about vaccines remains a major public health concern because it can weaken trust in proven disease-prevention measures and shape real-world health decisions. This concern is especially urgent for the MMR vaccine, which protects against measles, mumps, and rubella. Although MMR vaccination is highly effective, misleading claims about vaccine safety continue to circulate online and contribute to hesitancy, delayed vaccination, and outbreaks in under-vaccinated communities (CDC, 1999; The College of Physicians of Philadelphia, 2026; MacDonald, 2015; Gowda & Dempsey, 2013; Mathis et al., 2025).

Social media platforms have adopted several strategies to reduce misinformation, including labels, fact-checks, and crowdsourced correction systems. On X, Community Notes allow users to add corrective context to potentially misleading posts, with notes becoming visible only after being rated helpful by contributors with diverse perspectives (Community Notes, n.d.-c; Wojcik et al., 2022). This model is intended to reduce misinformation while preserving user participation, making it especially relevant to the tension between limiting harmful content and maintaining platform engagement.

However, most evaluations of Community Notes focus on the annotated *post* itself, such as whether reposts, likes, replies, or views decline after a note appears (Chuai et al., 2024; Slaughter et al., 2025), or on corrected *posters* (Kim et al., 2025). Less is known about downstream *viewer* behavior: what users do after encountering a note on someone else's post, whether their later posts contain misinformation, and whether short-term responses could affect broader network diffusion. This gap is important because corrective interventions may appear successful at the post level while still producing viewer-level responses that shape misinformation spread.

This study addresses that gap by using observational X data to compare users' MMR-related posting behavior before and after apparent exposure to MMR-related Community Notes. It also

extends the user-level analysis with a network diffusion simulation grounded in social contagion theory to explore how observed behavioral patterns could scale through a larger social network.

Specifically, this study aims to address the following research questions:

- RQ1. Is exposure to a post with an MMR-related Community Note associated with changes in MMR-related posting activity?
- RQ2. Is exposure to a post with an MMR-related Community Note associated with changes in the proportion of misinformation in MMR-related posts?
- RQ3. How might the user-level patterns found in RQ2 shape the diffusion of MMR-related misinformation at the network level?

Literature Review

MMR Misinformation as a Public Health Problem

Online misinformation has become a major social problem because it can deepen polarization, disrupt public policy, threaten public health, affect democratic processes, and create real-world safety risks (Ueta et al., 2025; Ecker et al., 2024; Pennycook et al., 2020; Roozenbeek et al., 2020; Bär & Feuerriegel, 2023). Social media intensifies these risks by allowing misinformation to spread quickly and form large cascades, especially in high-stakes areas such as politics and health (Ueta et al., 2025; Juul & Ugander, 2021; Kozyreva et al., 2023). As a result, researchers and policymakers have called for platforms to identify, implement, and evaluate interventions that reduce misinformation spread, including labeling, content moderation, crowdsourced annotations, and other user-level countermeasures (Calo et al., 2021; Donovan, 2020; Kozyreva et al., 2024; Wojcik et al., 2022).

Vaccine misinformation is especially consequential because it can affect real health decisions. Measles is a highly contagious respiratory infection that can cause serious complications, including pneumonia, encephalitis, dehydration, ear infections, and irreversible vision loss; more than 90% of susceptible people exposed to an infected person may develop measles (CDC, 1999). The MMR vaccine is highly effective, providing 96% protection against measles, 95% against mumps, and 94% against rubella (The College of Physicians of Philadelphia, 2026). After widespread MMR rollout, U.S. measles cases fell by more than 99%, contributing to measles elimination in 2000 (The College of Physicians of Philadelphia, 2026). However, misinformation linking MMR to autism has contributed to parental fear, vaccine delays, and refusal, increasing the number of susceptible individuals (MacDonald, 2015; Gowda, 2013). These gaps are dangerous because herd immunity requires about 95% coverage (The College of Physicians of Philadelphia, 2026). Recent outbreaks show that this remains a current concern: from January 1 to April 17, 2025, 800 U.S. measles cases were reported, 82% were linked to an outbreak in low-coverage communities, 85 patients were hospitalized, and three died (Mathis et al., 2025).

The Engagement-Misinformation Tradeoff

Although platforms face pressure to reduce misinformation, their business models create a competing incentive to maintain engagement. Social media platforms often depend on advertising revenue, which is tied to keeping users active for longer periods of time (Milli et al., 2025). This creates a challenge because misinformation can perform well in engagement-based systems. Edelson et al. (2021) found that publishers known for misinformation received about

six times more likes, shares, and interactions on Facebook than trustworthy news sources during the 2020 election period. Internal Facebook documents reported by The Wall Street Journal also suggest that Facebook resisted some fixes because of concerns that they would reduce user interaction (The Wall Street Journal, 2021). Therefore, an effective misinformation intervention would ideally reduce engagement with misleading content without reducing overall user engagement.

Community Notes as a Platform Intervention

Community Notes is one platform-level response to this challenge. On X, Community Notes allows contributors to add contextual information to posts that may be misleading, while other contributors rate whether those notes are helpful. Notes only appear publicly when they are rated helpful by contributors with diverse perspectives, rather than by simple majority rule. X describes this system as a collaborative, transparent approach in which users write and rate notes, while X does not choose which notes appear unless they violate platform rules (Community Notes, n.d.-c).

Prior research generally finds that Community Notes and related corrective features can reduce belief in or engagement with misleading content. Experimental studies show that Community Notes and similar annotations can improve users' understanding of misleading claims, increase trust in corrections, and reduce willingness to like, reshare, believe, or spread misleading content (Wojcik et al., 2022; Drolsbach et al., 2024; Kankham & Hou, 2024). Field studies provide similar evidence at the post level. Chuai et al. (2024) found that Community Notes reduced the spread of misleading posts by an average of 61.4%, while Slaughter et al. (2025) found significant post-note reductions in reposts, likes, replies, and views. However, these studies mainly evaluate the annotated posts themselves, often focusing on whether engagement with the corrected post decreases after the note appears.

The Need to Study Viewers and Diffusion

This post-level focus leaves important downstream effects less understood. Community Notes may reduce engagement with the original misleading post, but viewers who encounter a note may still respond in ways that shape later discussion and misinformation spread. This matters because platform interventions do not operate only at the individual or post level; their effects may also depend on how users respond and how those responses move through networks. Furutani et al. (2026) similarly cautions that even interventions that reduce individual susceptibility may have limited network-level effects unless they are strong, early, and large-scale enough to disrupt diffusion dynamics.

Some work examines real user behavior, but it often focuses on users whose own posts were corrected. Kim et al. (2025), for example, found that individual fact-checking replies could push corrected posters back into information bubbles, while collectively verified Community Notes softened that retreat. This shows that the form of correction matters, but it still centers on corrected posters rather than the broader audience exposed to notes.

Ueta et al. (2025) provides the clearest viewer-centered model. Studying HPV vaccine Community Notes in Japan, Ueta et al. compared users' posts before and after they appeared to view a note. They found that 73% of users maintained their stance after exposure, while posting

activity increased immediately after notes appeared, especially among vaccine-opposing users. This suggests that Community Notes may have more complex effects than simple correction: notes may reduce engagement with annotated posts while also triggering new discussion or resistance among some viewers.

Building on this gap, the present study examines downstream viewer behavior after exposure to MMR-related Community Notes. It extends Ueta et al.'s before-and-after framework in three ways: it shifts the topic from HPV to MMR misinformation, measures misinformation sharing rather than stance alone by identifying support for pre-fact-checked false or misleading claims, and adds a network-level simulation grounded in social contagion theory. This approach allows the study to assess whether short-term viewer responses to Community Notes may influence broader misinformation diffusion.

Methodology

Part I: User-Level

Dataset. This study follows a data collection procedure adapted from prior work on user responses to Community Notes (Ueta et al., 2025). I used X's publicly available Community Notes dataset (Community Notes, n.d.-a), specifically the "Notes" and "Note Status History" files, downloaded on March 19, 2026. The "Notes" file contains Community Note text and associated post IDs, while "Note Status History" provides note evaluation metadata, including helpfulness status and status timestamps (Community Notes, n.d.-b). The dataset spans from the earliest available note on January 28, 2021, through data available approximately 48 hours before collection.

I retained entries with non-empty note text and source post IDs, then filtered for MMR-related notes using the keywords "MMR," "MMR vaccine," "MMR vaccination," "measles vaccine," and "measles vaccination." To focus on notes that became and remained publicly visible, I kept only notes whose `firstNonNMRStatus` and `currentStatus` were both `CURRENTLY_RATED_HELPFUL`. If multiple qualifying notes appeared on the same post, I retained the earliest visible note. After manual review, I removed one irrelevant "MMR" case referring to matchmaking rating in gaming. The final Community Notes sample included 46 unique posts.

Following prior work, exposure to a Community Note was operationalized through user replies occurring after the note became visible. Using the timestamp of first visibility, I identified all unique users who replied to each post after the associated Community Note appeared. For each of these users, I then collected their full history of MMR-related posts, including replies and quote reposts, based on the same keyword set as above. To focus on within-user change, I restricted the sample to users who posted at least once in both the before and after periods. This ensures that the analysis captures changes in behavior among users already engaged with MMR-related content. Users who only posted after exposure were excluded because their activity may reflect first exposure to the topic via the post itself rather than a response to the Community Note, making it difficult to distinguish responses associated with the note from responses associated with the original post. Users who only posted before exposure were also excluded, as their lack of post-exposure activity prevents the measurement of any change over time. This procedure yielded a dataset of 95 unique users and 549 MMR-related posts.

Misinformation Classification. To identify misinformation in MMR-related posts, I used a claim-matching and stance-classification approach summarized in Figure 1.

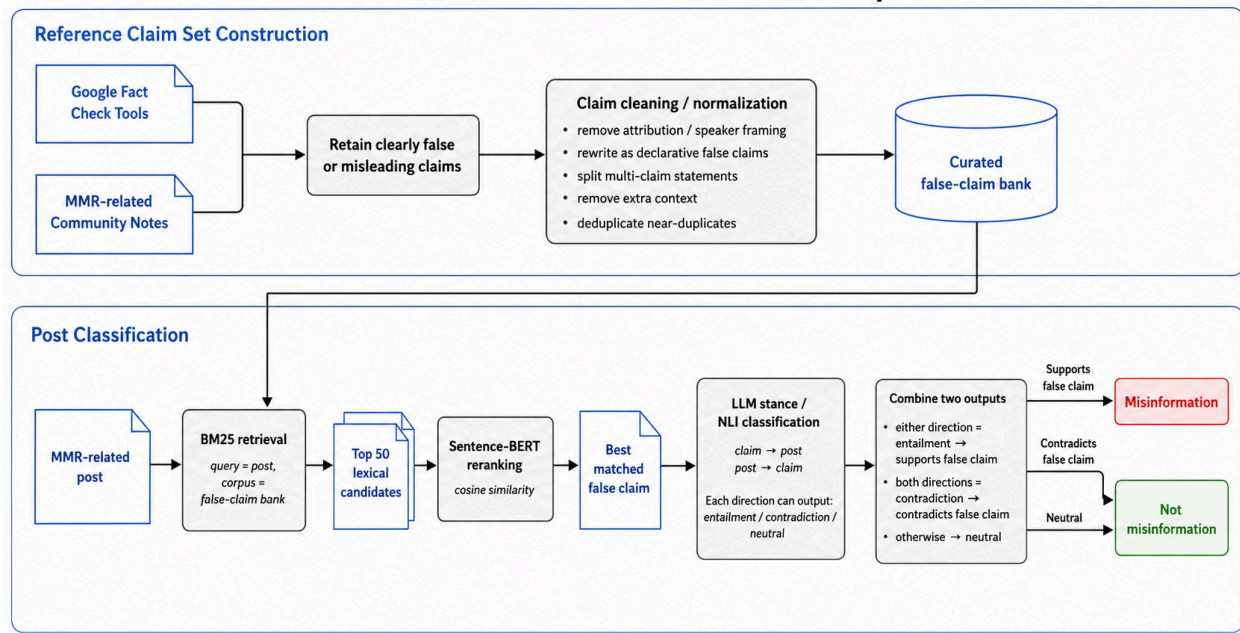


Figure 1: MMR misinformation classification pipeline.

False MMR claims were curated and cleaned into a claim bank, matched to posts using BM25 and Sentence-BERT, then classified with bidirectional LLM stance detection. Posts supporting a false claim were labeled misinformation.

I first curated a reference set of pre-fact-checked false or misleading MMR claims from Google Fact Check Tools using the keywords “MMR,” “MMR vaccine,” “MMR vaccination,” “measles vaccine,” and “measles vaccination” (Google, n.d.), as well as from the Community Notes in my dataset. This follows prior misinformation detection work that uses known false claims as the reference set for identifying whether social media posts are discussing or spreading misinformation (Hossain et al., 2020; Weinzierl & Harabagiu, 2022; Choi & Ferrara, 2023). I only retained claims that were clearly labeled as false or misleading, consistent with prior work that used claims that were “were unequivocally labeled false, incorrect, or fake” (Choi & Ferrara, 2023). Prior work simplifies fact-checked claims into clearer misinformation expressions by rephrasing longer statements into positive expressions of the misinformation (Hossain et al., 2020) and removing subject or speaker framing from claims (Jiang et al., 2022). Following this approach, I cleaned my MMR-related claims by removing attribution or speaker framing, such as changing “A parent on X claimed that the MMR vaccine causes autism” to “The MMR vaccine causes autism”; reducing long claims into clearer declarative statements, such as changing “There are concerns online that children may develop autism after receiving the MMR vaccine” to “The MMR vaccine causes autism”; splitting multi-claim statements, such as separating “The MMR vaccine causes autism and weakens children’s immune systems” into two claims; removing unnecessary context, such as deleting extra background about who posted the claim or when it circulated; and deleting duplicates or near duplicates, such as combining “MMR causes autism” and “The MMR vaccine leads to autism” into one claim.

Next, I matched each MMR-related post to the most relevant false claim by treating posts as queries and the curated false claims as the document corpus. For each post, I first used BM25 to retrieve up to the top 50 lexically similar claim candidates, then re-ranked those candidates using Sentence-BERT cosine similarity, selecting the highest-scoring match as the final post-claim pair. This follows claim-matching approaches that combine token-based retrieval with semantic similarity (Choi & Ferrara, 2023; Jiang et al., 2022).

Finally, I used an LLM to classify the relationship between each post and its matched false claim as entailment, contradiction, or neutral, following prior work that frames misinformation classification as a stance or natural language inference task (Hossain et al., 2020; Weinzierl & Harabagiu, 2022; Choi & Ferrara, 2023). Figure 2 provides examples of each classification label. In practice, this meant prompting the LLM to determine whether the tweet supported the false claim (entailment), rejected the false claim (contradiction), or was related but did not take a clear stance (neutral). I classified each pair twice, once in the claim-to-post direction and once in the post-to-claim direction, because entailment can depend on presentation order (Choi & Ferrara, 2023). I then combined the two outputs using the following rule: if either direction produced entailment, the post was labeled as supporting the false claim; if both directions produced contradiction, the post was labeled as contradicting the false claim; otherwise, the post was labeled neutral. Posts classified as entailment were treated as misinformation because they supported or repeated a pre-fact-checked false or misleading claim.

Tweet: “Parents should think twice about the MMR shot because it has been linked to autism.” Misconception: “The MMR vaccine causes autism.” Label: Entailment	Tweet: “The MMR vaccine has been studied for years, and there is no evidence that it causes autism.” Misconception: “The MMR vaccine causes autism.” Label: Contradiction	Tweet: “Our pediatric clinic is offering MMR vaccinations this Saturday for children who need them.” Misconception: “The MMR vaccine causes autism.” Label: Neutral
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Figure 2: Example LLM stance-classification labels.

Examples show how MMR-related posts were classified in relation to a matched false claim. Entailment indicates the post supports the false claim and was labeled misinformation; contradiction indicates the post rejects the false claim; neutral indicates the post does not take a clear stance toward the claim.

Part II: Network-Level Simulation

To examine how the user-level patterns observed in Part I might shape misinformation diffusion at the network level, I built a discrete-time network diffusion simulation grounded in the social contagion framework described above (Christakis & Fowler, 2013). This approach follows Furutani et al.’s (2026) broader simulation framework, which connects micro-level user

interventions to macro-level misinformation prevalence through network-based diffusion modeling. In my simulation, misinformation adoption was modeled as a function of interpersonal exposure, a baseline contagion rate, and each user's empirically derived susceptibility to misinformation.

The simulation used an epidemic-style Susceptible-Infected model adapted for repeated contact. Nodes represented users, and edges represented social connections. Time moved in daily event-time steps from 62 days before exposure to 62 days after exposure, consistent with prior work using approximately two-month pre-post windows to capture X user behavioral change (Ueta et al., 2025; Kim et al., 2025). Once a node adopted misinformation, it remained active and could repeatedly attempt to influence its neighbors on each later day.

The network was modeled as a Watts-Strogatz small-world graph with 15,000 nodes, an average degree of 10, and a rewiring probability of 0.1. I used this structure because small-world networks capture both local clustering and short path lengths, which are common features of social systems. Because I did not have access to users' actual follower or interaction networks, the graph was undirected and simulated rather than directly observed.

To ground the simulation in my empirical data, I assigned user attributes by sampling with replacement from the 95 observed users in the X dataset, allowing the larger simulated network to preserve the observed distribution of misinformation behavior. Following Furutani et al.'s (2026) use of user-level misinformation scores as a proxy for susceptibility, each simulated user received a baseline susceptibility score based on their overall misinformation proportion across the full 124-day observation window, calculated as the number of misinformation posts divided by the total number of MMR-related posts.

I also created a user-specific event-time spike parameter from the Part I dataset. This parameter was included because the Part I event-time analysis motivated a closer examination of day 0 patterns; the corresponding empirical result is reported in the Part I results section. For each observed user, I calculated their average daily misinformation proportion during the pre-exposure period, using days -62 to -1 and coding zero-post days as 0. I then calculated the user's misinformation proportion on day 0 and divided it by the pre-exposure daily baseline to create a user-specific spike multiplier. In the simulation, this multiplier was applied only on day 0.

I compared two scenarios: a No-Note world, representing a counterfactual scenario where users were not exposed to a Community Note, and a Note world, representing the observed exposure scenario. In the No-Note world, users' susceptibility remained constant over time, and no event-time spike parameter was introduced. In the Note world, users had the same overall susceptibility, but the day 0 spike multiplier derived from the Part I event-time analysis was applied. This design allowed me to examine how a temporary event-time change in misinformation behavior could shape diffusion while holding longer-term susceptibility constant.

At each time step, every active node attempted to influence each inactive neighbor. The probability that an inactive neighbor adopted misinformation was calculated as: contagion

rate $\times$ user susceptibility $\times$ spike multiplier, adapting Furutani et al.'s (2026) formulation of transmission as a function of misinformation contagiousness and receiver susceptibility. The contagion rate was set to 0.026, based on Furutani et al.'s empirically calibrated estimate of misinformation contagiousness. Adoption was permanent, so once a user adopted misinformation, they remained active for the rest of the simulation. Initial adopters were selected as the top 3% of users with the highest baseline susceptibility.

Finally, I repeated the simulation across multiple runs to reduce dependence on any single random network or sample of user attributes. Each run generated a new network structure, resampled user attributes from the observed data, and simulated both the No-Note and Note worlds using the same initial conditions. The simulation ran in batches until the average final prevalence stabilized.

Results

Part I: User-Level

RQ1. RQ1 tested whether users' MMR-related posting activity changed after exposure to a post with a visible MMR-related Community Note. In the paired analysis, I compared each of the 95 users' total number of MMR-related posts before exposure with that same user's total number of MMR-related posts after exposure. On average, users posted 2.863 times before exposure and 2.916 times after exposure. The median number of posts was 2.0 in both periods. The mean within-user change was 0.053 posts, indicating a very small average increase after exposure. A paired-samples t-test showed that the within-user difference was not statistically significant, $t = 0.145$, $p = .885$. As a robustness check, I also ran a Wilcoxon signed-rank test, which similarly showed no statistically significant within-user difference, $W = 1262.5$, $p = .771$. These results indicate that exposure to a post with a visible MMR-related Community Note was not associated with a significant overall change in users' MMR-related posting activity.

I also examined how MMR-related posting activity was distributed over event time, measured by relative day from exposure. This analysis used the user-day as the unit of analysis and calculated the average number of MMR-related posts per user per day from 62 days before exposure to 62 days after exposure. This allowed me to assess whether posting activity changed temporarily around the moment of exposure, even if the overall pre/post comparison did not show a significant difference. As shown in Figure 3, posting activity appeared to peak on day 0, the day of exposure. The mean baseline daily posting rate was 0.0194 posts per user per day, compared with 0.1875 posts per user on day 0. To assess whether this spike was statistically significant, I estimated a negative binomial regression predicting posts per user-day, with an indicator for day 0, controls for temporal trends, and standard errors clustered at the user level. The model showed a significant increase on day 0 ($\beta = 1.200$, $p = .003$), indicating that users posted about 3.3 times as frequently on the exposure day as on other days in the observation window (95% CI [1.52, 7.25]). Thus, although the full-window analysis found no significant change in total MMR-related posting after exposure, the event-time analysis revealed a brief surge in activity concentrated on day 0.

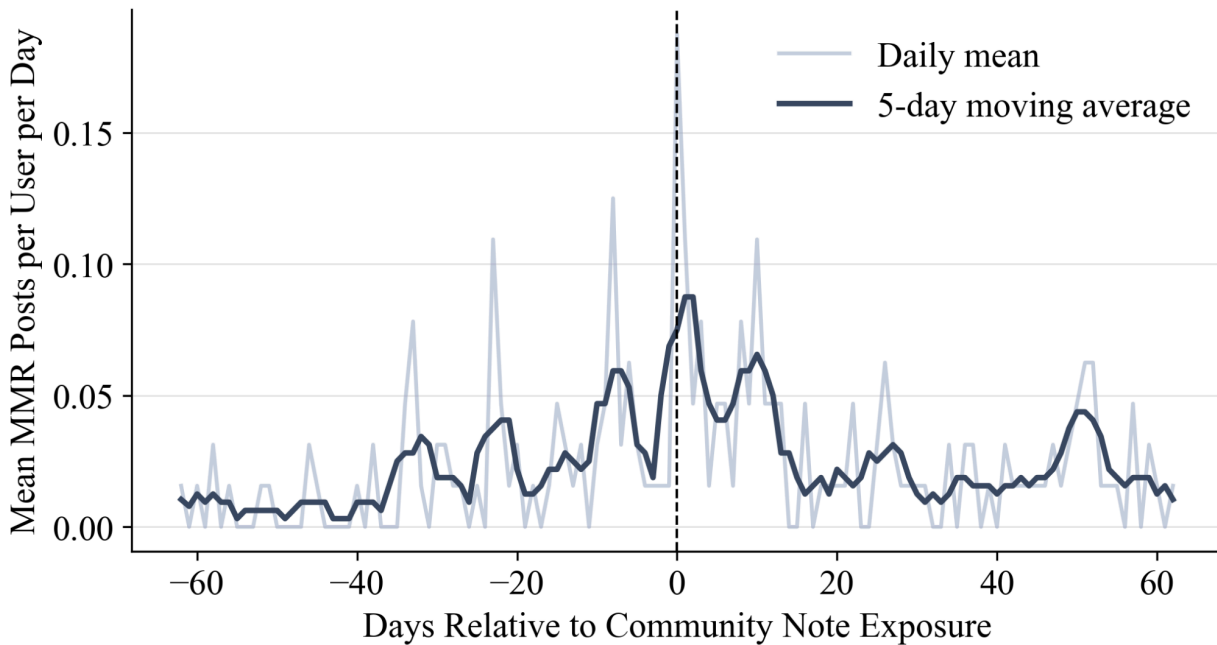


Figure 3: MMR-related posting activity around Community Note exposure.

The figure shows the average number of MMR-related posts per user per day from 62 days before to 62 days after exposure to a post with a visible MMR-related Community Note. Day 0 represents the exposure day.

RQ2. RQ2 tested whether users' proportion of MMR-related misinformation changed after exposure to a post with a visible MMR-related Community Note. Using the same paired within-user approach as RQ1, I compared each user's misinformation proportion before and after exposure. On average, users' misinformation proportion was 0.218 before exposure and 0.214 after exposure. The median misinformation proportion was 0.0 in both periods. The mean within-user change was -0.004, indicating a very small average decrease after exposure. A paired-samples t-test showed that the within-user difference was not statistically significant, $t = -0.118$, $p = .906$. As a robustness check, I also ran a Wilcoxon signed-rank test, which similarly showed no statistically significant within-user difference, $W = 586.5$, $p = .988$. These results indicate that exposure to a post with a visible MMR-related Community Note was not associated with a significant overall change in users' proportion of MMR-related misinformation.

I then used the same event-time approach to examine daily misinformation proportion around exposure. As in RQ1, the day-level analysis used the user-day as the unit of analysis across the 62-day pre/post window. Similar to the RQ1 posting activity results, as shown in Figure 4, misinformation proportion seemed to spike on day 0. The mean baseline misinformation proportion was 0.0045 per user per day, compared with 0.0521 on day 0. The mean difference between day 0 misinformation proportion and baseline daily misinformation proportion was 0.0476. I tested this increase using a binomial regression model of the share of posts that were misinformation per user-day, weighted by the total number of posts and adjusted for temporal trends, with standard errors clustered by user. The model indicated a statistically significant day 0 increase ($\beta = 2.454$, $p = .009$). The corresponding odds ratio was 11.63 (95% CI [1.83, 73.88]), suggesting that, among active user-days, posts were substantially more likely to contain

misinformation on the day of exposure than on other days in the window. However, this estimate is based on a relatively small number of posts on day 0, resulting in a wide confidence interval. Therefore, while the direction of the effect is clear, the magnitude should be interpreted with caution. Overall, the event-time analysis indicates a short-term increase in misinformation proportion on day 0, although the precision of the estimate is limited.

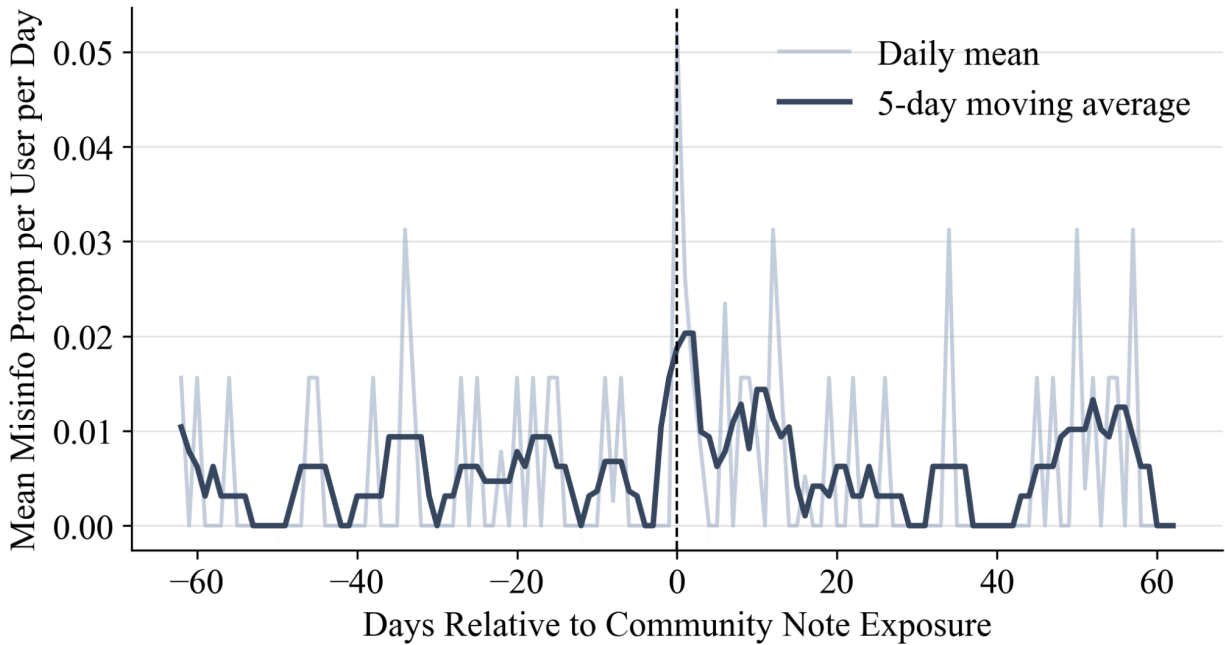


Figure 4: Misinformation proportion around Community Note exposure.

The figure shows the average proportion of MMR-related posts classified as misinformation per user per day from 62 days before to 62 days after exposure to a post with a visible MMR-related Community Note. Day 0 represents the exposure day.

Part II: Network-Level Simulation

RQ3. RQ3 looked into how the user-level patterns identified in RQ2 might shape the diffusion of MMR-related misinformation at the network level. The simulation compared two scenarios: a No-Note world, where users' susceptibility remained constant and no event-time spike was introduced, and a Note world, where the day 0 spike parameter derived from the user-level analysis was applied. This modeling choice was based on the Part I results: the full-window analyses showed no significant overall change in misinformation proportion after exposure, while the event-time analyses showed a short-term day 0 increase in misinformation proportion. Therefore, the Note world was designed to model the main pattern observed in the user-level data: little evidence of sustained behavioral change, but a temporary day 0 event-time spike.

Across simulation runs, the Note world produced more misinformation diffusion than the No-Note world. Final misinformation prevalence increased from 0.3831 in the No-Note world to 0.3966 in the Note world, a mean difference of 0.0134. This difference was statistically significant, $t = 10.764$, $p < .001$, with the Wilcoxon robustness check showing the same pattern. The clearest difference appeared in peak daily spread, or the highest number of new adopters on a single day. In the No-Note world, peak daily spread was 49.42 new adopters, while in the Note

world it increased to 274.13 new adopters. This means that adding the day 0 spike parameter corresponded to about 224.72 additional new adopters at the fastest point of diffusion, $t = 87.227$, $p < .001$. This early difference carried forward into total diffusion volume. The No-Note world produced an average of 5296.93 total adoption events, compared with 5498.42 in the Note world, or about 201.48 additional adoption events overall. This difference was also statistically significant, $t = 10.764$, $p < .001$, with the Wilcoxon robustness check again producing a significant result. The gap between the two scenarios also persisted beyond the immediate event window. As shown in Figure 5, the two scenarios begin to diverge at day 0, with the Note world maintaining a higher cumulative number of misinformation adopters throughout the remainder of the simulation period. During days 0-5, the average gap between the Note and No-Note worlds was 232.03 adopters. During days 30 and after, the average gap remained 236.46 adopters, and the final gap was 201.48 adopters. The late-period gap was significantly greater than zero, $t = 12.782$, $p < .001$. Therefore, the simulation results show that introducing the day 0 spike parameter increased both the speed and reach of misinformation diffusion. The largest change occurred in peak daily spread, but the difference also persisted into later time periods and resulted in higher final prevalence in the Note world.

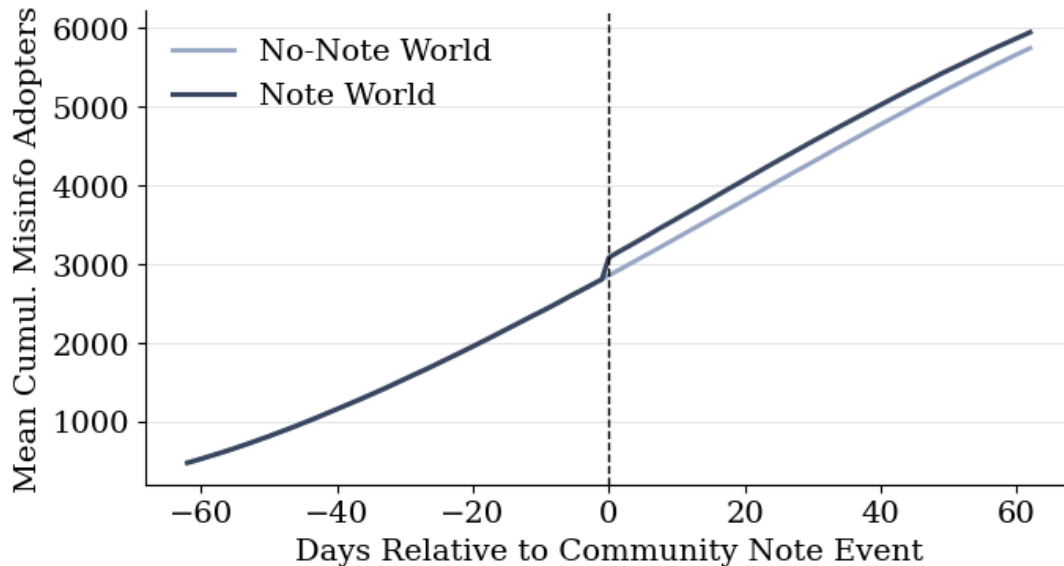


Figure 5: Mean cumulative misinformation adopters in the Note and No-Note worlds. The dashed vertical line marks day 0, the Community Note event. The Note world applies the day 0 spike parameter derived from the user-level analysis; the No-Note world does not.

Discussion

Part I: User-Level

At the user level, the key pattern is that Community Note exposure was not associated with sustained behavioral change, but it did coincide with a short-term spike on the day of exposure. This spike is important because it involved both higher posting activity and a higher misinformation proportion. In other words, users were not only posting more after encountering the note; a larger share of what they posted supported false or misleading MMR claims. This suggests that some exposed users may have responded to the correction by posting against it and reasserting the misinformation. This interpretation is consistent with Ueta et al. (2025), who found that HPV-related Community Notes did not substantially change most users' stances but

did trigger an immediate increase in posting, especially among vaccine-opposing users. Because most of the notes in their study supported vaccination, Ueta et al. suggested that opposing users may have posted more in response to information that challenged their position. My findings support and extend that interpretation: the day 0 increase was not only an engagement spike, but also a misinformation spike, suggesting that corrective context may prompt some viewers to actively defend or amplify the misleading claims being corrected.

Psychological reactance theory helps explain why this kind of response might occur. Psychological reactance theory argues that when people perceive their freedom to think, believe, or express something as being challenged, they may respond by reasserting that freedom, sometimes through oppositional behavior (Brehm, 1966). In this context, a visible Community Note may be interpreted by some users as a challenge to their view of the original post. The result may be a short-term defensive response: instead of disengaging from the misinformation, some users may become more motivated to post in favor of it.

The timing of the effect is consistent with this interpretation. Because reactance is typically situational and temporary, it would be expected to appear most strongly when the corrective stimulus is most visible or salient. The day 0 increases in both posting activity and misinformation proportion fit that pattern. However, this study cannot directly measure users' beliefs, motivations, or perceptions of the Community Note, so reactance should be treated as a plausible interpretation of the observed behavior rather than direct evidence of users' psychological states.

Part II: Network-Level Simulation

The network-level simulation shows why the short-term user-level spike matters. The original intervention goal was to reduce misinformation while maintaining engagement. However, the user-level results showed the opposite pattern on the day of exposure: engagement increased, but misinformation increased with it. When this temporary spike was introduced into the simulation, it increased both the speed and reach of misinformation diffusion. This suggests that the day 0 response did not simply create more discussion; it created more opportunities for misinformation to spread.

This pattern is consistent with the structure of the diffusion model. In the Note world, the day 0 spike temporarily increased the probability that exposed users would adopt misinformation. Once those additional users adopted it, they became new spreaders in later time steps, allowing a short-lived response to produce a small but persistent increase in overall diffusion. Because the simulation returned susceptibility to baseline after day 0, the network-level difference was driven by the timing of the spike rather than by any assumed permanent change in user behavior.

This is practically meaningful in the context of an intervention intended to reduce misinformation spread. Community Notes are designed to add corrective context to misleading posts, but in this case, exposure was associated with a short-term increase in both engagement and misinformation sharing. The simulation suggests that this response could modestly amplify misinformation diffusion, meaning the intervention may have partially reversed, or backfired against, its intended effect. This shows that evaluations of Community Notes focused only on annotated posts may miss downstream viewer responses that shape wider diffusion.

Limitations and Future Work

This study has several limitations. First, the real-world X dataset was relatively small, consisting of 95 users and 549 MMR-related posts. This was partly due to the narrow scope of the topic and the filtering procedure required for the analysis. After identifying users who replied to posts with visible MMR-related Community Notes, I limited the sample to users who had keyword-matched MMR posts both before and after exposure. This supported the within-user design but substantially reduced the final sample size. Second, exposure to Community Notes was inferred from reply behavior. Following prior work, I treated users who replied to a post after a Community Note became visible as users who were likely exposed to the note. However, replying does not guarantee that a user actually read the note. At the same time, this approach excludes users who may have viewed the note but did not reply. As a result, the sample likely overrepresents more active or engaged users and may not capture the behavior of less active viewers. Third, misinformation classification relied on an LLM-based claim-matching and entailment classification process. Although this approach allowed misinformation labels to be applied systematically across the dataset, LLMs are not perfectly accurate and may misclassify posts that involve sarcasm, ambiguity, indirect references, or missing conversational context. Future work should validate these classifications against human-coded labels or compare multiple model prompts and classifiers. Finally, while the findings are consistent with psychological reactance theory, this study cannot directly test psychological reactance. The analysis observes posting behavior, not users' beliefs, motivations, or emotional responses to Community Notes. Therefore, the reactance interpretation should be treated as a plausible explanation for the observed short-term posting spike rather than direct evidence of reactance. Future research could pair behavioral data with surveys, interviews, or experimental designs to test whether users actually perceive Community Notes as threatening or corrective in ways that trigger oppositional responses.

Ethical Statement

This study analyzes publicly available X posts and Community Notes data. I did not interact with users, alter platform activity, or intervene in any conversations. Because vaccine misinformation is a sensitive public health topic and social media posts may contain identifiable usernames or contextual details, I report findings only in aggregate and avoid quoting individual posts or reproducing user-identifying information. The study also relies on automated misinformation classification using claim matching and an LLM-based entailment approach, so labels should be understood as analytical classifications rather than definitive judgments about individual users. Overall, the risk of direct harm of this study is low, but the study still carries limitations related to privacy, classifier error, and the sensitive nature of vaccine misinformation research.

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